

# EQUIFLOW QA ASSESSMENT

AFTER REPORT: 100% Test Case Pass Rate & Production Deployment

STATUS: ' 19/19 PASSED (MongoDB Atlas Cloud & Render Live Host)

Generated: 2026-08-15 | Live Host: <https://tactive.onrender.com/>

## 1. Executive Resolution & Architecture Summary

All injected regressions and test assertion mismatches have been resolved. The database was successfully migrated from SQLite to MongoDB Atlas Cloud Cluster. The entire stack has been deployed live to Render at <https://tactive.onrender.com/> with 100% passing automated test coverage.

## 2. AFTER Test Execution Matrix (100% PASS)

| Category          | Test Scenario & Expectation                              | After Outcome     |
|-------------------|----------------------------------------------------------|-------------------|
| 1. Happy Path     | GET /api/assets -> Seeded inventory list returned        | PASS [HTTP 200] ' |
| 1. Happy Path     | GET /api/health -> Health status check                   | PASS [HTTP 200] ' |
| 1. Happy Path     | VIP User auto-confirmed reservation (No Admin queue)     | PASS [HTTP 201] ' |
| 1. Happy Path     | Standard User reservation -> Enters PENDING queue        | PASS [HTTP 201] ' |
| 1. Happy Path     | Admin can approve PENDING reservation                    | PASS [HTTP 200] ' |
| 1. Happy Path     | Admin can reject PENDING reservation                     | PASS [HTTP 200] ' |
| 1. Happy Path     | User can cancel own PENDING reservation                  | PASS [HTTP 200] ' |
| 2. Business Rules | CONFLICT: Overlapping dates rejected (conflictEngine)    | PASS [HTTP 409] ' |
| 2. Business Rules | QUOTA: Standard user blocked at >3 active bookings       | PASS [HTTP 400] ' |
| 2. Business Rules | BLACKOUT: Booking blocked during admin maintenance       | PASS [HTTP 409] ' |
| 2. Business Rules | PENALTY (VIP): 3 days overdue -> \$45 (50% VIP discount) | PASS [HTTP 200] ' |
| 2. Business Rules | PENALTY (Standard): On-time return = \$0 penalty         | PASS [HTTP 200] ' |
| 3. Validation     | REJECT: Inverted dates (start > end)                     | PASS [HTTP 409] ' |
| 3. Validation     | REJECT: Booking duration > 14 days limit                 | PASS [HTTP 409] ' |
| 3. Validation     | REJECT: Non-existent asset ID returns 404                | PASS [HTTP 404] ' |
| 3. Validation     | REJECT: Unauthenticated request returns 401              | PASS [HTTP 401] ' |
| 3. Validation     | REJECT: Standard user cannot create blackouts            | PASS [HTTP 403] ' |
| 3. Validation     | REJECT: Missing required fields (Zod 400)                | PASS [HTTP 400] ' |
| 3. Validation     | REJECT: Invalid date format (Zod 400)                    | PASS [HTTP 400] ' |

### 3. Captured Terminal Output (100% Green Run)

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